

Project Report – Leave Approval System

1. INTRODUCTION

1.1 Project Overview

- 2 Many organizations still rely on manual methods such as emails, paper forms, or informal communication to manage employee leave requests. This often leads to delays in approvals, lack of transparency, and difficulty in tracking the status of leave requests.
- 3 To address these issues, I designed and developed the **Leave Approval System using the ServiceNow platform**. The system digitizes the entire leave management process by providing a centralized workflow where employees can submit leave requests online, and managers can review and take appropriate action efficiently.
- 4 The application follows a **role-based access control model**, ensuring secure and structured access for Employees, Managers, and Administrators. Employees can create and track their leave requests, Managers are responsible for approving or rejecting requests, and Administrators handle system configuration and manage user roles and access control.
- 5 The workflow is automated using **Flow Designer**, which triggers email notifications whenever a leave request is approved or rejected. This improves communication, reduces manual effort, and ensures timely updates to users.
- 6 Overall, this system enhances transparency, reduces paperwork, and streamlines the leave approval process within an organization.

1.2 Purpose

The purpose of this project is to replace traditional manual leave request handling with a centralized and automated digital workflow built on the ServiceNow platform.

Key Objectives:

- Provide a centralized platform for employees to submit leave requests
- Automate the approval workflow using ServiceNow Flow Designer
- Enable Managers to approve or reject employee leave requests
- Allow Administrators to manage roles, access control, and system configuration
- Implement role-based access control for Employees, Managers, and Administrators
- Send email notifications on leave request status updates (Approved/Rejected)
- Reduce manual effort and improve process transparency
- Demonstrate practical implementation of ServiceNow concepts such as:
 - Scoped Applications
 - Custom Tables
 - Business Rules
 - UI Policies
 - Roles & Access Control
 - Flow Designer & Notifications

Overall, the system reduces paperwork, improves transparency, and enhances communication between employees and managers.

To demonstrate the practical implementation of ServiceNow application development concepts, including scoped applications, custom tables, Business Rules, Flow Designer, UI Policies, roles, and notifications.

2. PROJECT OVERVIEW

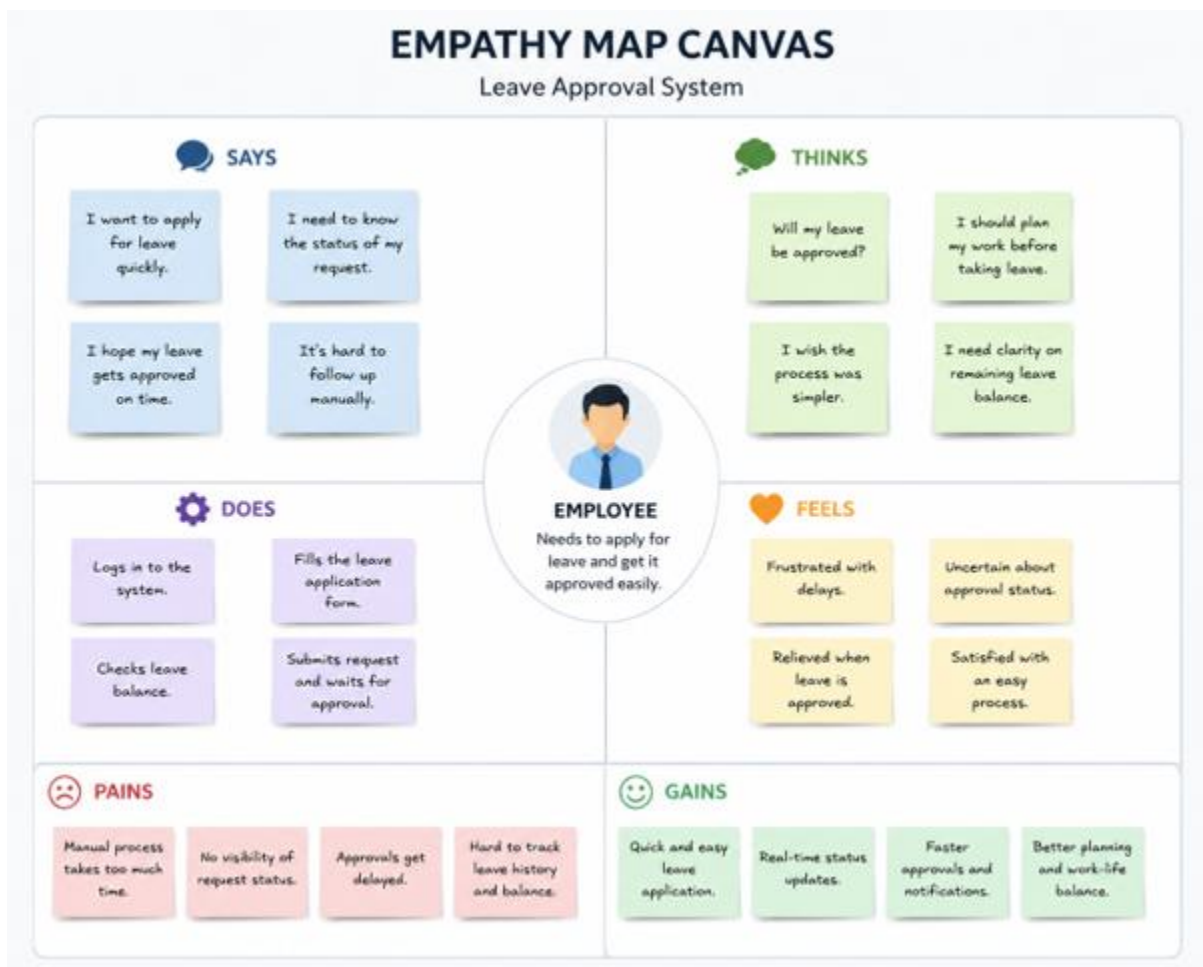
2.1 Problem Statement

#	Section	Description	Key Issues/Emotions
1	I AM 	An employee who needs to apply for leave for personal, medical, urgent, or other valid reasons.	Leave need Workload Approval delay Uncertainty
2	I'M TRYING TO 	Submit a leave request easily and know whether it is approved or rejected in time.	Speed Status transparency Time sensitivity Planning work
3	BUT 	The leave process is manual and involves multiple steps like forms, emails, and follow-ups. I do not know the current status or who is approving my request.	Manual process Multiple approvals Lack of visibility Status ambiguity Delays
4	BECAUSE 	Leave requests are managed through paper forms, emails, or spreadsheets. There is no centralized system, so requests can be missed, approvals take time, and data is not well organized.	Paper-based / Email-based No central system Fragmented communication Approval bottlenecks Data inconsistency
5	WHICH MAKES ME FEEL 	Frustrated, stressed, and uncertain about leave approval. It becomes difficult to plan my work and manage responsibilities.	Frustration Stress Uncertainty Work disruption Lack of trust in the process

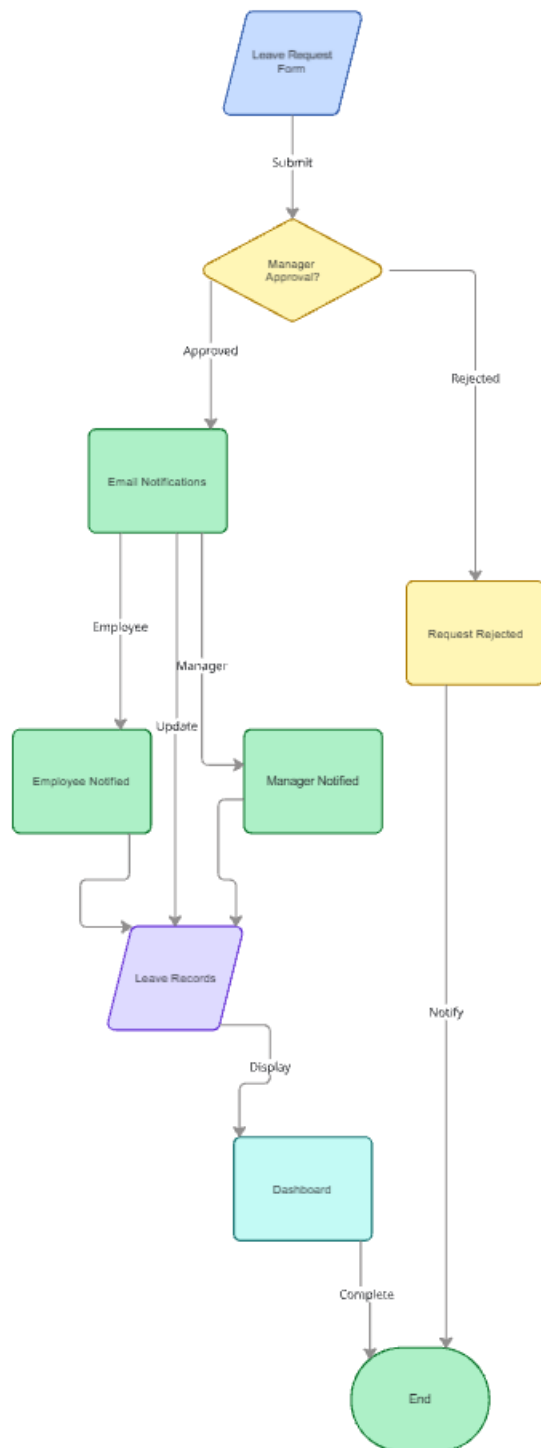
Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Employee	Submit a leave request quickly and track its approval status.	The manual leave process is slow and difficult to monitor.	It requires paperwork or multiple email communications, and	Frustrated, Uncertain, and Stressed.

				approvals may be delayed.	
PS-2	Manager / Approver	Review and approve employee leave requests efficiently.	Managing multiple leave requests manually takes time and increases the chance of errors.	Leave requests come from different employees and need proper verification before approval.	Overloaded, Delayed, and Less Productive.

2.2 Empathy Map Canvas

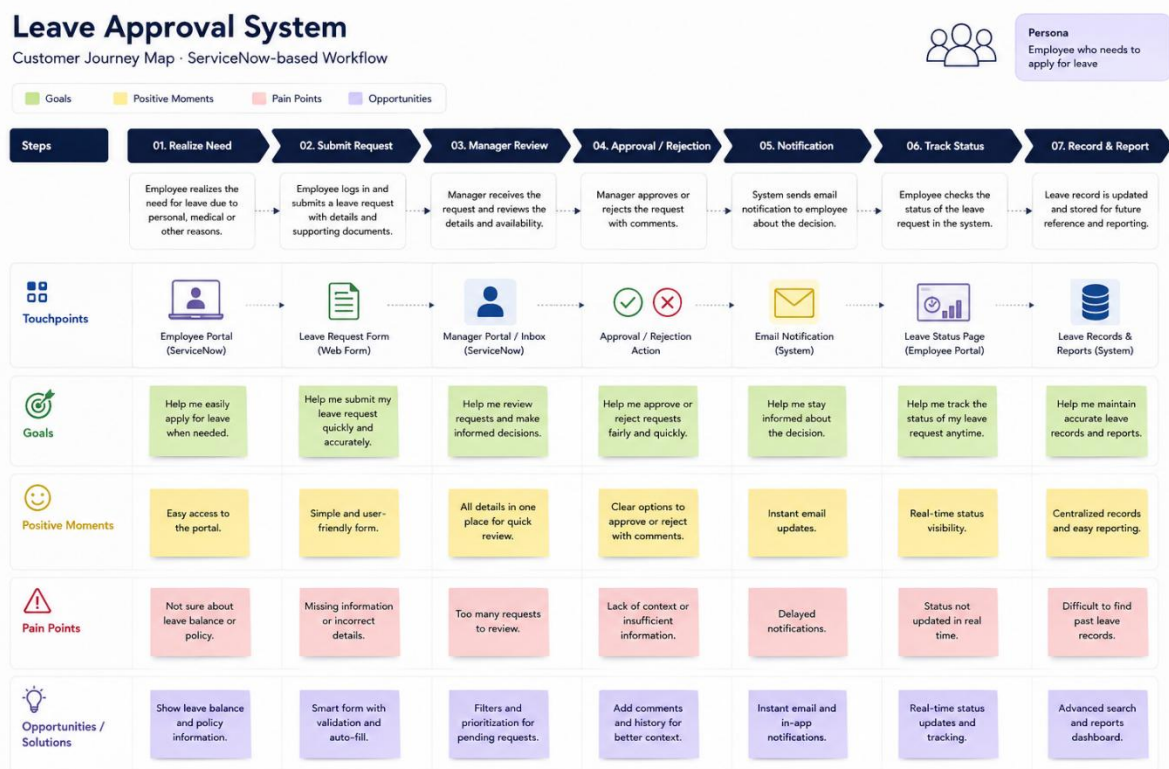


2.3 Brainstorming



3. REQUIREMENT ANALYSYS

3.1 Customer Journey Map



3.2 Solution Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication & Role Management	Authenticate users using the existing ServiceNow Users (sys_user) table. Assign Employee role to employees. Assign Manager role to approvers. Assign Administrator role for system administration.
FR-2	Leave Request Management	Employees can create and submit leave requests. Employees can enter leave type, leave dates, reason, and attach supporting documents (if required).
FR-3	Leave Request Review and Response	Managers can view pending leave requests. Approve leave requests. Reject leave requests with comments. Update the leave request status.
FR-4	Leave Status Tracking	Employees can view the status of submitted leave requests. Status includes Pending , Approved , and Rejected .
FR-5	Workflow Automation & Notifications	Trigger a ServiceNow Flow Designer workflow after leave submission and approval/rejection. Send approval email notification. Send rejection email notification. Update leave records automatically.

Non-functional Requirements:

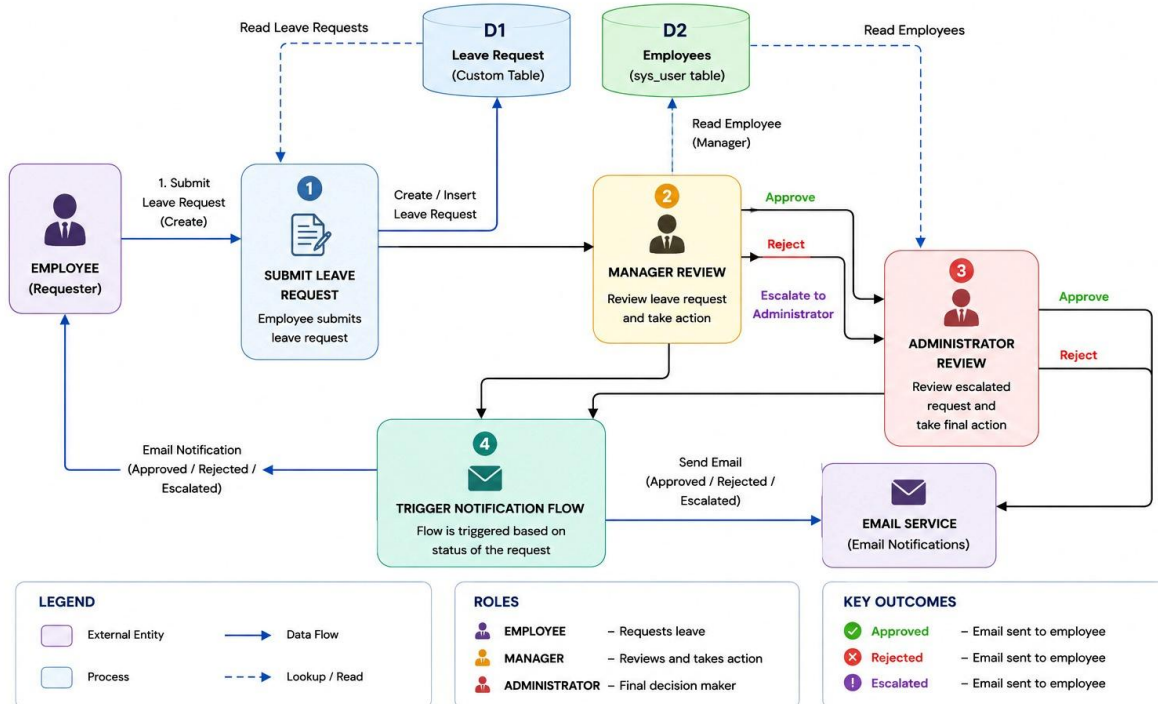
Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application shall provide a simple, user-friendly interface for employees, managers, and administrators.
NFR-2	Security	Access shall be controlled using ServiceNow roles. Employees can access only their own leave requests, while managers and administrators have access according to their assigned roles.
NFR-3	Reliability	The application shall accurately store leave requests and process approvals without data loss.
NFR-4	Performance	Leave requests, approvals, and email notifications shall be processed with minimal delay under normal usage.
NFR-5	Availability	The Leave Approval System shall be available whenever the ServiceNow instance is operational.
NFR-6	Scalability	The application shall support an increasing number of employees, managers, and leave requests without major design changes.

3.3 Data Flow Diagram: A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DATA FLOW DIAGRAM (DFD LEVEL 1)

Leave Approval System (Employees Only)



3.4 Technology Stack

Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Employee, Manager, and Administrator views for leave management	ServiceNow Workspace / Forms / Lists
2	Application Logic	Leave request creation, validation, approval, and status management	Business Rules, Client Scripts
3	Workflow Automation	Leave approval workflow,	Flow Designer

		notifications, and record updates	
4	Database	Stores employee details and leave requests	sys_user, Custom Leave Request table
5	Email Service	Sends approval and rejection email notifications	ServiceNow Notifications
6	Security	Role-based access for employees, managers, and administrators	ACLs, Employee Role, Manager Role, Admin Role
7	Infrastructure	ACLs, Employee Role, Manager Role, Admin Role	ServiceNow Cloud Platform

Table 2: Application Characteristics

Characteristic	Description	Technology
Security	Role-based access ensures employees, managers, and administrators access only authorized data.	ACLs, ServiceNow Roles
Scalability	Supports a growing number of employees and leave requests without major changes.	ServiceNow Platform
Availability	The application is available whenever the ServiceNow instance is operational.	ServiceNow Cloud
Performance	Leave requests, approvals, and notifications are processed efficiently.	Flow Designer, Business Rules
Reliability	Leave records and approval history are stored securely with accurate processing.	ServiceNow Database
Maintainability	Forms, workflows, business rules, and notifications can be updated easily without affecting existing data.	ServiceNow Studio, Flow Designer

4. PROJECT DESIGN

4.1 Problem Solution Fit

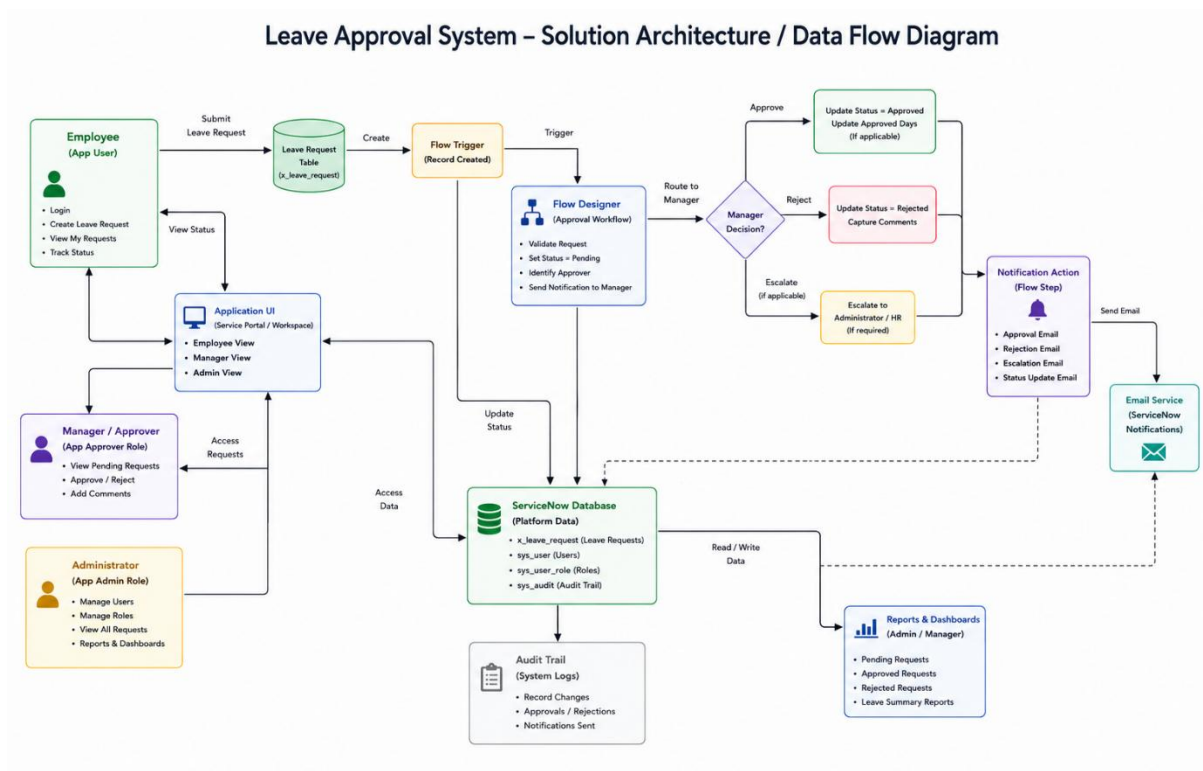
Section	Content
1. Customer Segment(s)	• Employees • Managers/Approvers • System Administrators
2. Jobs-to-be-done / Problems	• Submit leave requests online • Get quick approval or rejection • Track leave request status • Reduce paperwork • Maintain centralized leave records
3. Triggers	• Employee needs leave for personal, medical, vacation, or emergency reasons • Manager receives a new leave request • Administrator manages employee records and workflows
4. Emotions (Before / After)	Before: Frustrated, uncertain, stressed about approval delays. After: Confident, informed, satisfied, and stress-free.
5. Available Solutions	• Paper leave forms • Email requests • Phone calls • Manual approval process • Spreadsheet-based leave tracking
6. Customer Constraints	• Manual approval delays • No centralized leave tracking • Employees unaware of request status • Difficult to maintain leave records • Communication through multiple channels
7. Behaviour	• Employees submit leave requests via paper or email • Follow up repeatedly with managers • Managers manually review requests • Employees wait for approval updates
8. Channels of Behaviour	Online: ServiceNow application, Email notifications. Offline: Direct communication with managers or HR.
9. Problem Root Cause	The organization lacks a centralized digital leave management system. Manual leave processing causes delays, poor visibility, inconsistent tracking, and increased administrative effort.
10. Your Solution	Develop a ServiceNow Leave Approval System where employees submit leave requests online. Managers review and approve or reject requests. The system automatically updates the leave status, sends email notifications

Section	Content
	using ServiceNow Notifications, securely stores all leave records, and allows employees to track the status of their requests through the application.

4.2 Proposed Solution

S.No	Parameter	Description
1	Problem Statement (Problem to be solved)	Many organizations manage employee leave requests manually through paper forms, emails, or verbal communication. This results in delayed approvals, lack of transparency, difficulty in tracking leave status, and inefficient communication between employees and managers.
2	Idea / Solution Description	Develop a ServiceNow-based Leave Approval System where employees submit leave requests digitally. Managers review and approve or reject the requests through the application. ServiceNow Flow Designer automatically updates the request status and sends email notifications to employees. Employees can also track the status of their leave requests through the application.
3	Novelty / Uniqueness	The solution integrates role-based access control, automated leave approval workflows, email notifications, centralized leave record management, and separate Employee, Manager, and Administrator views within a single ServiceNow application. It eliminates manual paperwork and improves the efficiency of the leave approval process.
4	Social Impact / Customer Satisfaction	The application improves transparency, reduces paperwork, speeds up leave approvals, and enhances communication between employees and managers. Employees receive timely updates, while managers can efficiently manage leave requests through a centralized platform.
5	Business Model (Revenue Model)	The solution is designed for organizations as an internal leave management application. It can be deployed across companies using the ServiceNow platform. Future enhancements can support a Software-as-a-Service (SaaS) model for multiple organizations.
6	Scalability of the Solution	The application is built on the ServiceNow platform and can scale to support multiple departments, branches, and thousands of employees. Additional modules such as leave balance management, HR integration, reporting dashboards, mobile access, and analytics can be added without significant architectural changes.

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

Product Backlog, Sprint Schedule, and Estimation

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	8	2 Days	22 June 2026	23 June 2026	8	23 June 2026
Sprint-2	9	2 Days	24 June 2026	25 June 2026	9	25 June 2026
Sprint-3	8	2 Days	26 June 2026	27 June 2026	8	27 June 2026
Sprint-4	5	1 Day	28 June 2026	28 June 2026	5	28 June 2026

Velocity:

- Total Story Points: 30
- Total Duration: 7 days

Average Velocity = $30 \div 7 \approx 4.3$ story points/day

6. FUNCTIONAL AND PERFORMANCE TESTING

2. Milestone 1 – Application Setup

Objective

Create the scoped application and initialize the project.

Activities Performed

- Created a new scoped application.
- Configured the application menu and modules.
- Prepared the development environment.

CREATE APP

Let's get started on your new app.

Add a name and description that define the purpose of your app. You can also add a thumbnail image.

Name *

Description

Scope * ☒ Scoped ☐ Global

Browse or drag to upload

BMP, GIF, ICO, JPEG, JPG, PNG, SVG

Cancel Continue

Enhance your productivity by using AI-powered Now Assist. Contact your admin and ask them to set it up. [Learn more about using Now Assist](#)

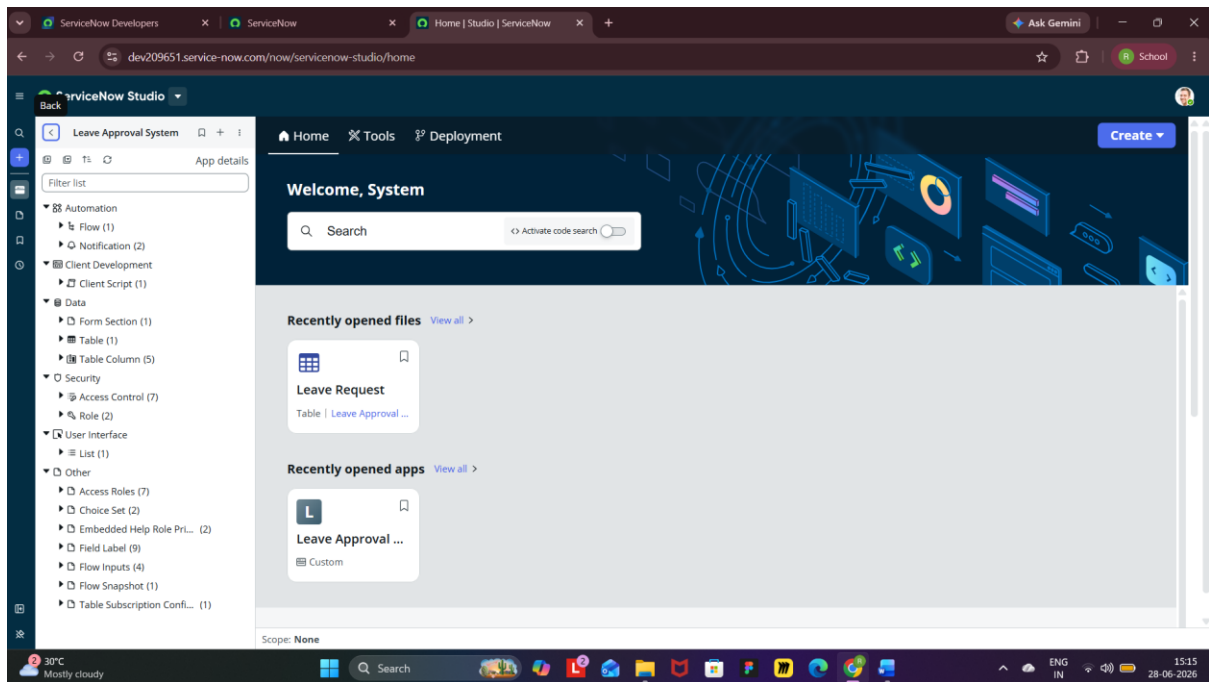
3. Milestone 2 – Database Configuration

Objective

Create the Leave Request table for storing leave applications.

Activities Performed

- Designed the Leave Request table.
- Added fields such as Employee, Leave Type, Start Date, End Date, Reason, Status, Manager Comments, and Approval Date.
- Configured reference fields to the ServiceNow Users table (sys_user).

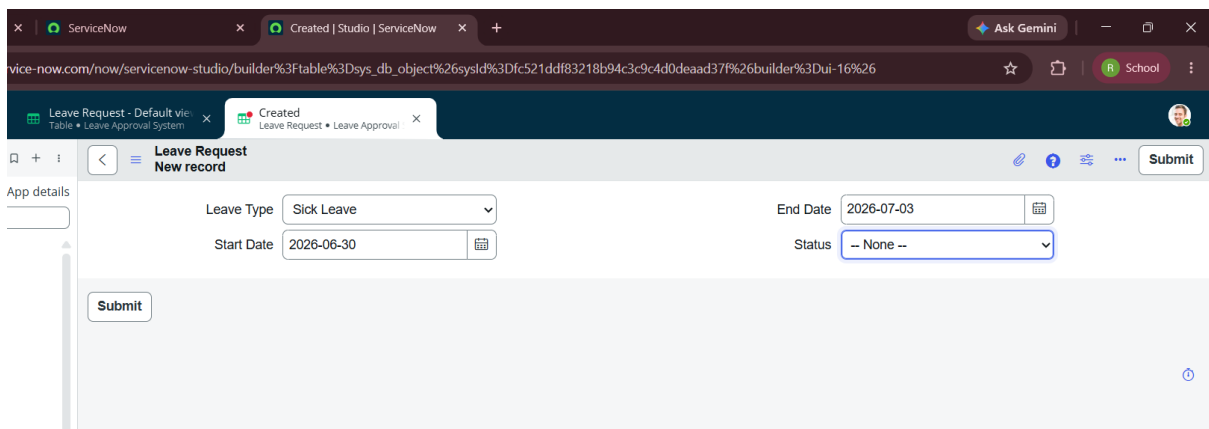


4. Milestone 3 – User & Role Configuration

Configure users and application roles.

Activities Performed

- Created Employee user accounts.
- Created Manager user account.
- Assigned **Employee** role to employees.
- Assigned **Manager** and **Administrator** roles to authorized users.



5. Milestone 4 – User Interface Development

Objective

Develop user-friendly interfaces for students and faculty.

Activities Performed

- Designed the Employee Leave Request form.
- Configured separate Employee and Manager view.
- Implemented UI Policies for dynamic form behavior.

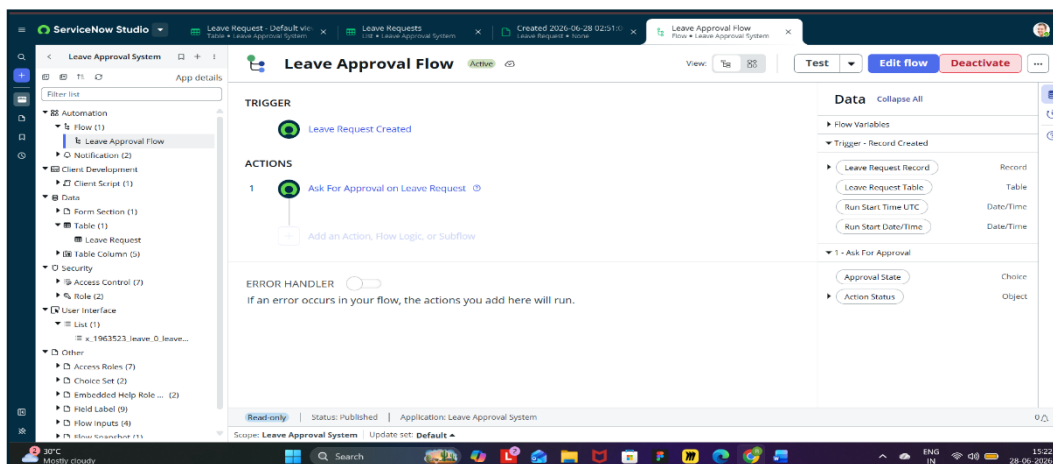
6. Milestone 5 – Workflow Automation

Objective

Automate leave approval and notification process.

Activities Performed

- Developed a Flow Designer workflow.
- Triggered the workflow whenever the leave status changes.
- Configured email notifications for Approved and Rejected leave requests.
- Automatically updated leave request status.



Notification Preview

Preview records for this breakdown source

Event Creator

System Administrator

Preview Record

Created 2026-06-28 02:51:06

Users

Abel Tuter

Subject

Leave Request Approved

Body

Your leave request has been approved.

[Unsubscribe](#) | [Notification Preferences](#)

Notification Preview

Preview records for this breakdown source

Event Creator

System Administrator

Preview Record

Created 2026-06-28 02:51:06

Users

Abel Tuter

Subject

Leave Request Approved

Body

Your leave request has been approved.

[Unsubscribe](#) | [Notification Preferences](#)

7. Milestone 6 – Application Testing

Objective

Validate the end-to-end functionality of the application.

Test Scenario 1

Employee submits a leave request.

Expected Result:

Leave request is created successfully.

Status:

Passed

The screenshot shows the ServiceNow interface for 'Leave Requests'. The table contains one record with the following details:

Created	Created by	Updated	Updated by	Updates	End Date	Leave Type	Start Date	Status	Tags
2026-06-28 02:51:06	admin	2026-06-28 02:51:06	admin	0	2026-07-03	Sick Leave	2026-06-30	Approved	Add tag...

Test Scenario 2

Manager approves a leave request.

Expected Result:

Status changes to **Approved** and an email notification is sent to the employee.

Status:

Passed

The screenshot shows the ServiceNow interface for 'Leave Requests'. The table contains two records with the following details:

Created	Created by	Updated	Updated by	Updates	End Date	Leave Type	Start Date	Status	Tags
2026-06-28 03:00:30	admin	2026-06-28 03:00:30	admin	0	2026-07-01	Casual Leave	2026-06-23	Approved	Add tag...
2026-06-28 02:51:06	admin	2026-06-28 02:51:06	admin	0	2026-07-03	Sick Leave	2026-06-30	Approved	Add tag...

Test Scenario 3

Manager rejects a leave request.

Expected Result:

Status changes to **Rejected** and an email notification is sent to the employee.

Status:

Passed

Created	Created by	Updated	Updated by	Updates	End Date	Leave Type	Start Date	Status	Tags
2026-06-28 03:01:19	admin	2026-06-28 03:01:19	admin	0	2026-07-03	Earned Leave	2026-06-24	Approved	Add tag...
2026-06-28 03:00:30	admin	2026-06-28 03:00:30	admin	0	2026-07-01	Casual Leave	2026-06-23	Approved	Add tag...
2026-06-28 02:51:06	admin	2026-06-28 02:51:06	admin	0	2026-07-03	Sick Leave	2026-06-30	Approved	Add tag...

Test Scenario 4

Role-based access testing.

Expected Result:

Employees can access only their own leave requests, while managers can view and manage assigned requests.

Status:

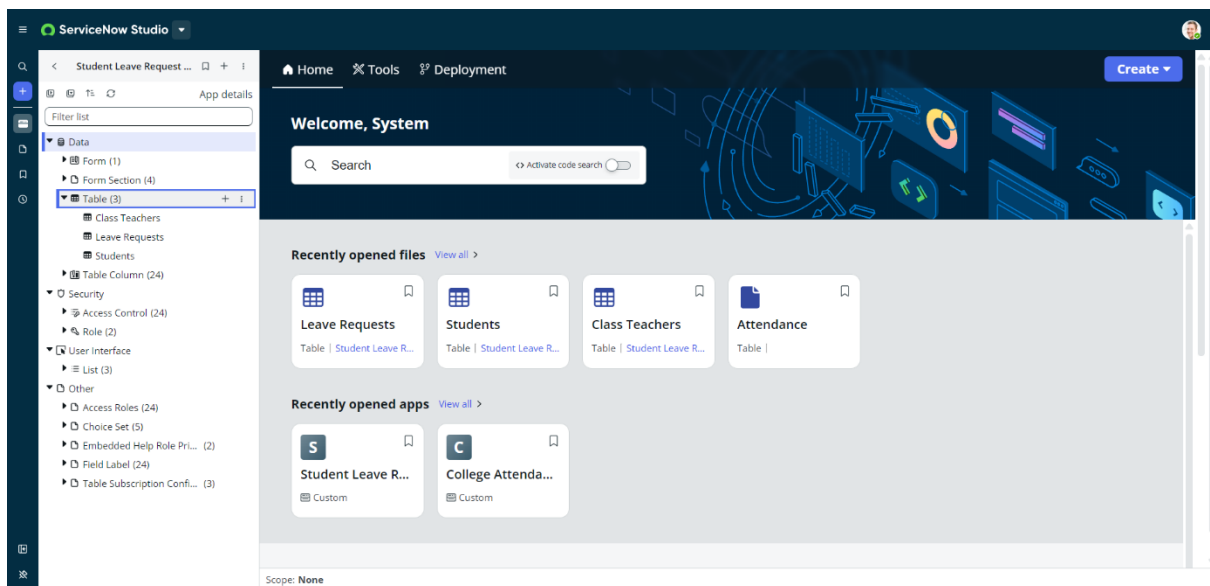
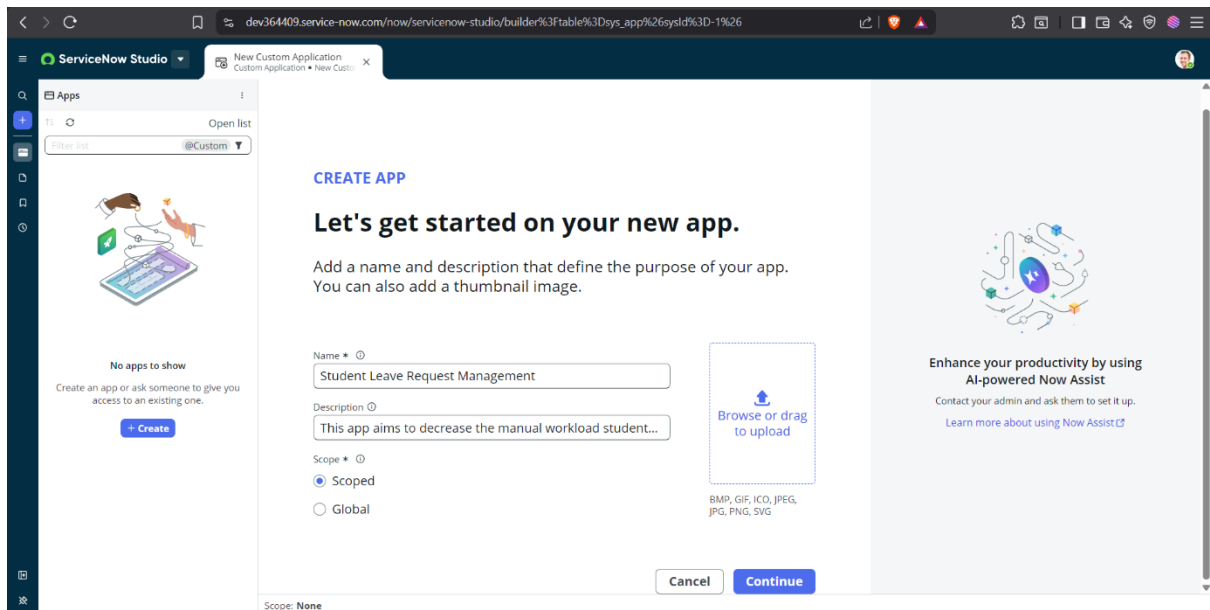
Passed

Created	Created by	Updated	Updated by	Updates	End Date	Leave Type	Start Date	Status	Tags
2026-06-28 03:01:19	admin	2026-06-28 03:01:19	admin	0	2026-07-03	Earned Leave	2026-06-24	Approved	Add tag...
2026-06-28 03:01:57	admin	2026-06-28 03:01:57	admin	0	2026-07-11	Work From Home	2026-07-09	Approved	Add tag...
2026-06-28 03:00:30	admin	2026-06-28 03:00:30	admin	0	2026-07-01	Casual Leave	2026-06-23	Approved	Add tag...
2026-06-28 02:51:06	admin	2026-06-28 02:51:06	admin	0	2026-07-03	Sick Leave	2026-06-30	Approved	Add tag...

8 Development Summary

The **Leave Approval System** successfully implements a centralized leave management workflow using the ServiceNow platform. The application provides secure role-based access, employee leave request submission, manager approval and rejection, workflow automation using Flow Designer, email notifications, and centralized leave record management. All functional requirements were successfully implemented and validated through testing.

7. RESULTS



UserG Vijay Kumar

Update

Set Password

Delete

User IDG.Vijay.Kumar

First nameG Vijay

Last nameKumar

Title

Department

Password needs reset

Locked out

Active

Identity typeHuman

Internal Integration User

Emailvijaykumar@gmail.com

Language-- None --

Calendar integrationOutlook

Time zoneSystem (America/Los_Angeles)

Date formatSystem (2026-08-19)

Business phone

Mobile phone

PhotoClick to add...

Update

Set Password

Delete

Related Links

[View linked accounts](#)

[View Subscriptions](#)

[Reset a password](#)

[\[SN Utils\] Versions \(0\)](#)

Roles (2)

Groups (1)

Delegates

User Client Certificates

UserChinna Rao

Update

Set Password

Delete

User IDChinna.Rao

First nameChinna

Last nameRao

Title

Department

Password needs reset

Locked out

Active

Identity typeHuman

Internal Integration User

Emailchinna@gmail.com

Language-- None --

Calendar integrationOutlook

Time zoneSystem (America/Los_Angeles)

Date formatSystem (2026-08-19)

Business phone

Mobile phone

PhotoClick to add...

Update

Set Password

Delete

Related Links

[View linked accounts](#)

[View Subscriptions](#)

[Reset a password](#)

[\[SN Utils\] Versions \(0\)](#)

Roles (2)

Groups (1)

Delegates

User Client Certificates

User

Vijay Krishna

Update

Set Password

Delete

User ID

Vijay.Krishna

Email

vk@gmail.com

First name

Vijay

Language

-- None --

Last name

Krishna

Calendar integration

Outlook

Title

Time zone

System (America/Los_Angeles)

Department

Date format

System (2026-08-19)

Password needs reset

☐

Business phone

Locked out

☐

Mobile phone

Active

☒

Photo

Click to add...

Identity type

Human

Internal Integration User

☐

Update

Set Password

Delete

Related Links

[View linked accounts](#)

[View Subscriptions](#)

[Reset a password](#)

[\[SN Utils\] Versions \(0\)](#)

Roles (2)

Groups (1)

Delegates

User Client Certificates

User

G Vijay Kumar

Update

Set Password

Delete

Active

☒

Photo

Click to add...

Identity type

Human

Internal Integration User

☐

Update

Set Password

Delete

Related Links

[View linked accounts](#)

[View Subscriptions](#)

[Reset a password](#)

[\[SN Utils\] Versions \(0\)](#)

Roles (2)

Groups (1)

Delegates

User Client Certificates

Role

Search

Actions on selected rows...

Edit...

User = G Vijay Kumar

☐	Role	State	Inherited	Inheritance Count
☐	snc_required_script_writer_permission	Active	true	1
☐	x_1963518_studen_0_admin	Active	false	

<<

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1

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23

dev364409.service-now.com/now/nav/ui/classic/params/target/sys_ui_policy_action.do%3Fsys_id%3D-1%26sys_is_list%3D1...

servicenow All Favorites History Workspaces UI Policy Action - New Record Application scope: Student Leave Request Management Update set: Default (Student Leave Request Manag...

UI Policy Make comment field mandatory when the class teacher escalates the leave application to hod View: Advanced

Table Leave Requests [x_1963518_studen_0_leave Application Student Leave Request Management

Active ☒

* Short description Make comment field mandatory when the class teacher escalates the leave application to hod

Order 100

When to Apply Script

Conditions Add Filter Condition Add OR Clause

Global ☒ On load ☒

Reverse if false ☒ Inherit ☐

Update Delete

Related Links

Convert this to Data Policy

Default view

Run Point Scan

UI Policy Actions (1) UI Policy Related List Actions

for text Search

Actions on selected rows... New

dev364409.service-now.com/now/workflow-studio/builder%3FtypeSysId%3D2d85e527439231106c4bb0117fb8f208%26sys...

Workflow Studio New Flow Flow: None

Let's get the details for your flow

Flow name * Email Notification for Students

Application * Student Leave Request Management

Description * This lets students know whether his/her leave requests are approved/escalated.

> Show additional properties

Cancel Build flow

ServiceNow Developers | Leave Requests | ServiceNow

dev364409.service-now.com/now/nav/ui/classic/params/target/x_1963518_studen_0_leave_requests_list.do%3Fsysparm_us...

serviceNow All Favorites History Workspaces Leave Requests Search

Leave Requests Name Search

All > Approve = false

Name	Regd Number	Email	From Date	To Date	Leave Reason	Description	Class Teacher	Approve	Approved by	Reject	Rejected by	Es
Chinna Rao	501	chinna@gmail.com	2026-06-27	2026-06-29	Health	I am suffering from fever and hence requ...	G Vijay Kumar	false	(empty)	false	(empty)	fak
Vijay Krishna	5b2	vk@gmail.com	2026-06-25	2026-07-10	Health	I am planning to go to a trip for 15 day...	G Vijay Kumar	false	(empty)	true	Ramesh Y	fak

1 to 2 of 2

18:32 27-06-2026

Preview Email

Dear Vijay Krishna,

Your Leave Request from 2026-07-01 to 2026-07-10 has been successfully approved by your Class Teacher and your attendance will be acknowledged

Ref:MSG0001704_H8s3mtVH30EGytMGaqMb

Preview Email

Dear Vijay Krishna,

Your leave request from 2026-07-01 to 2026-07-10 has been escalated to HOD level due to Leave is out of scope. Depends on HOD

Please stay tuned for further updates from the HOD.

Ref:MSG0001703_MLHfRYfa7QxV9ufjXJCE

8. ADVANTAGES AND DISADVANTAGES

Advantages

1. **Paperless Leave Management**

Eliminates paper forms and manual documentation by digitizing the entire leave request process.

2. **Centralized Leave Tracking**

All leave requests are stored in a single system, making it easy for students and faculty to access and manage records.

3. **Role-Based Access Control**

Students can access only their own leave requests, while faculty members can review and manage requests assigned to them.

4. **Improved Transparency**

Students can view the status of their leave requests, reducing uncertainty and the need for repeated follow-ups.

5. **Automated Email Notifications**

The system automatically sends email notifications when a leave request is approved, rejected, or escalated.

6. **Efficient Approval Workflow**

Class Teachers can process leave requests directly and manually escalate requests exceeding five days to the HOD when necessary.

7. **Reduced Administrative Workload**

Faculty members can manage leave requests through a centralized interface instead of handling requests via multiple communication channels.

8. **Secure User Management**

User authentication is handled through the existing ServiceNow Users (sys_user) table with application-specific roles.

9. **Scalable Architecture**

The application can be extended to support multiple departments, additional approval levels, or integration with other campus systems.

10. **Easy Maintenance**

Since the application is built on the ServiceNow platform using configurable components such as Flow Designer and Business Rules, future modifications can be implemented with minimal effort.

Disadvantages

1. Requires ServiceNow Platform

The application depends on a ServiceNow instance and cannot operate independently without the platform.

2. Internet Dependency

Users must have an active internet connection to submit, review, or approve leave requests.

3. Manual Escalation Required

Leave requests exceeding five days require the Class Teacher to manually forward them to the HOD, which relies on timely faculty action.

4. Limited to Institutional Workflow

The application is designed specifically for student leave management and requires customization for use in other domains.

5. Initial User Configuration

Student and faculty accounts, roles, and reference fields must be configured before the system can be used effectively.

6. Email Delivery Dependency

Notifications depend on proper email configuration within the ServiceNow instance.

7. No Attendance Integration

The current implementation does not automatically synchronize approved leave with the institution's attendance management system.

8. Limited Reporting Features

Basic reporting is supported, but advanced analytics and dashboards require further development.

9. CONCLUSION

The **Student Leave Automation System** successfully demonstrates how the ServiceNow platform can be utilized to digitize and streamline the student leave approval process within educational institutions. The application replaces traditional manual methods with a centralized, role-based system that enables students to submit leave requests online while allowing Class Teachers and Heads of Department (HODs) to efficiently review, approve, reject, or manually escalate requests based on institutional policies. Through the implementation of ServiceNow features such as scoped applications, custom tables, Business Rules, Flow Designer, UI Policies, roles, and email notifications, the system provides a structured and transparent workflow that improves communication and reduces administrative effort.

The project successfully achieved its primary objectives by providing a secure, user-friendly, and scalable leave management solution. Functional testing verified that all major features—including leave submission, approval workflow, manual escalation, status tracking, and automated email notifications—performed as expected. Overall, the proposed solution enhances operational efficiency, improves the user experience for both students and faculty, and serves as a strong foundation for future enhancements such as attendance integration, mobile access, analytics dashboards, and advanced reporting capabilities.

10. FUTURE SCOPE

The Student Leave Automation System provides a strong foundation for digital leave management and can be further enhanced with additional features to improve functionality and user experience. Future enhancements may include integration with the institution's attendance management system to automatically update attendance records after leave approval, eliminating manual intervention. The application can also be extended with a dedicated mobile interface, enabling students and faculty to submit and review leave requests from smartphones.

Additional improvements include implementing analytics dashboards to monitor leave trends, department-wise statistics, and approval turnaround times. The system can be integrated with academic calendars to validate leave requests against examination schedules and institutional holidays. Multi-level approval workflows can be introduced for larger institutions requiring approvals from additional authorities. Furthermore, SMS or mobile push notifications can complement email notifications for faster communication. These enhancements would make the application more comprehensive, scalable, and suitable for deployment across multiple educational institutions while continuing to improve administrative efficiency and the overall user experience.

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